Sangeon Jeon

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Education

Georgia Institute of Technology | Atlanta, GA
Master of Science in Electrical and Computer Engineering, GPA 4.00 /4.0 — January 2025 – December 2025(Expected)
Bachelor of Science in Electrical and Computer Engineering, GPA 4.00 /4.0 — January 2023 – December 2024

Korea Advanced Institute of Science and Technology | Daejeon, South Korea
Bachelor of Science in Electrical Engineering, GPA 3.95 /4.3 — February 2018 – February 2025

Work Experience

Dong Il Vision | Anyang, South Korea — *June – August 2023*
Intern / Software Engineer

- Developed a C# WPF labeling tool for computer vision dataset generation, contributing to a product differentiation.

Research

Kyung Hee Center for Digital Health | Seoul, South Korea (Advisor: Prof. Dong Keon Yon)
Global prevalence of age-related macular degeneration (AMD) — *Spring 2025 - Present*

- Led a Bayesian meta-analysis (PyMC5, adaptive Metropolis) using DisMod-MR, a hierarchical Bayesian regression tool developed by GBD, with covariates and regional variation across 104 studies to estimate global AMD prevalence (1990–2021).
- Utilized Expectation Maximization algorithm to separate disease stages from mixed reports.
- Migrated the public DisMod-MR codebase from PyMC2 to PyMC5, enabling modern Bayesian computation.

Global prevalence of obsessive-compulsive and related disorders (OCD) — *Summer 2025*

- Applied a Bayesian hierarchical framework (PyMC5, adaptive Metropolis sampling) to estimate OCD prevalence across 204 countries using GBD population data; Currently under review at JAMA Psychiatry.

SIPLAB @ Gatech| Atlanta, GA (Advisor: Prof. Christopher J. Rozell)
CLOC Unnatural Subspace System Identification — *Spring 2024*

- Conducted research on developing an in silico testbench for experiments controlling neural populations through optical simulation, incorporating RNN into an existing brain-record fitting method.
- Co-led a team of 7 researchers alongside a fellow leader, overseeing project strategy and delegating tasks effectively while contributing to advanced opsin dynamics modeling.

Project

Korea Advanced Institute of Science and Technology | Daejeon, South Korea
KAIST Maker-a-thon (Topic: Item for the Elderly Living Alone) — *Summer 2024*

- Participated in three weeks of development, brainstorming and prototyping an item suitable for the topic using materials and equipment provided by KAIST makerspace. Led a team of four as the leader and won the top award with $2000.
- Proposed an RFID token system to manage the expiration dates of food in refrigerators. Innovated the pre-existing RFID refrigerator system by installing an RFID reader on a token container instead of the refrigerator.

Georgia Institute of Technology | Atlanta, GA
Senior Design Project: Plant Physiology — *Spring 2024*

- Collaborated with Vivent (Switzerland) to develop a system recording and replaying Venus flytrap action potentials, successfully achieving trap closure via electrical stimulation.
- Designed system architecture and implemented Seeeduino–MATLAB communication with a custom GUI, enabling reliable signal acquisition and playback up to 100 Hz.

Awards

Roger Webb Award ECE Senior Scholar Awards (2025), 1st Prize at KAIST Maker-a-thon (2024), Dean's List (5 semesters)

Relevant Coursework

KAIST: Probability and Statistics, Linear Algebra, Introduction to Machine Learning
Gatech: Statistical Machine Learning, Graphical Models in Machine Learning, Digital Processing of Speech Signal

Skills

Programming: Python (Pytorch, PyMC), MATLAB, Excel VBA
Languages: Korean (native), English (professional), Japanese (intermediate)